

EXPERIMENT – 5

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Performing Various Configurations of IPTables Firewall in Kali Linux

Aim

To perform and analyze various firewall configurations using IPTables in Kali Linux to control network traffic and enhance system security.

Software Requirements

- **Host Operating System:** Windows / Linux / macOS
- **Virtualization Software:** Oracle VirtualBox
- **Guest Operating System:** Kali Linux
- **Firewall Tool:** IPTables
- **System Requirements:**
 - RAM: Minimum 4 GB (8 GB recommended)
 - Disk Space: Minimum 30 GB
 - Processor: Intel/AMD with Virtualization support

Introduction

IPTables is a powerful firewall utility in Linux used to configure rules for filtering incoming and outgoing network traffic. It operates by defining rules based on IP addresses, ports, and protocols. Configuring IPTables helps protect systems from unauthorized access, network attacks, and malicious traffic. Kali Linux provides built-in support for IPTables, making it suitable for learning firewall configurations in a controlled environment.

Procedure

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start the Kali Linux virtual machine.
3. Log in using Kali credentials.

Step 2: Open Terminal and Check IPTables

1. Open **Terminal** in Kali Linux.
2. Check IPTables version:
3. iptables --version

Step 3: View Existing Firewall Rules

1. Display current IPTables rules:
2. sudo iptables -L

Step 4: Set Default Policies

1. Set default policy to drop incoming traffic:
2. `sudo iptables -P INPUT DROP`
3. Allow outgoing traffic:
4. `sudo iptables -P OUTPUT ACCEPT`

Step 5: Allow Localhost Traffic

1. Allow loopback interface traffic:
2. `sudo iptables -A INPUT -i lo -j ACCEPT`

Step 6: Allow SSH Access

1. Allow SSH connections on port 22:
2. `sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT`

Step 7: Allow HTTP and HTTPS Traffic

1. Allow HTTP traffic:
2. `sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT`
3. Allow HTTPS traffic:
4. `sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT`

Step 8: Block a Specific IP Address

1. Block traffic from a specific IP:
2. `sudo iptables -A INPUT -s 192.168.1.100 -j DROP`

Step 9: Save IPTables Rules

1. Save the configured rules:
2. `sudo iptables-save > firewall_rules.txt`

Output:

View existing firewall rules:

```
kali@kali:~$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source                destination
DROP       all  --  192.168.1.102          anywhere
DROP       all  --  192.168.1.0/24         anywhere
DROP       tcp  --  anywhere              anywhere            tcp dpt:ssh
ACCEPT     tcp  --  192.168.1.102          anywhere            tcp dpt:ssh state NEW,ESTABLISHED

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
ACCEPT     tcp  --  anywhere              176.32.103.205
ACCEPT     tcp  --  anywhere              54.239.28.85
ACCEPT     tcp  --  anywhere              s3-console-us-standard.console.aws.amazon.com
DROP       tcp  --  anywhere              anywhere            tcp dpt:http
DROP       tcp  --  anywhere              anywhere            tcp dpt:https
kali@kali:~$
```

View existing firewall rules:

```
(plama@kali)-[~]
$ sudo iptables -S
-P INPUT ACCEPT
-P FORWARD ACCEPT
-P OUTPUT ACCEPT

(plama@kali)-[~]
$ sudo iptables -I INPUT 1 -p tcp --dport 80 -j DROP

(plama@kali)-[~]
$ sudo iptables -I INPUT 2 -s 192.168.200.128 -p tcp --dport 80 -j ACCEPT

(plama@kali)-[~]
$ sudo iptables -S
-P INPUT ACCEPT
-P FORWARD ACCEPT
-P OUTPUT ACCEPT
-A INPUT -p tcp -m tcp --dport 80 -j DROP
-A INPUT -s 192.168.200.128/32 -p tcp -m tcp --dport 80 -j ACCEPT

(plama@kali)-[~]
$
```

Allow localhost / loopback:

```
[root@host-192-168-1-10 ~]# iptables -I INPUT -i lo -j ACCEPT
[root@host-192-168-1-10 ~]# iptables -I OUTPUT -o eth1 -d 192.168.1.20 -j DROP
[root@host-192-168-1-10 ~]# service iptables status
Table: filter
Chain INPUT (policy ACCEPT)
num target      prot opt source                destination
1  ACCEPT        all  --  0.0.0.0/0              0.0.0.0/0
2  ACCEPT        all  --  192.168.1.20           0.0.0.0/0

Chain FORWARD (policy ACCEPT)
num target      prot opt source                destination

Chain OUTPUT (policy ACCEPT)
num target      prot opt source                destination
1  DROP          all  --  0.0.0.0/0              192.168.1.20

Chain RH-Firewall-1-INPUT (0 references)
num target      prot opt source                destination
```

- Block a specific IP
- Final rules / save rules

Result

Various firewall configurations were successfully implemented using IPTables in Kali Linux. The experiment demonstrates how firewall rules can be used to control network traffic and improve system security.

Reference

<https://www.kali.org/docs/networking/iptables/>

Note

This experiment is performed **only for academic and ethical purposes** in a controlled laboratory environment.